

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 2: Airfoil Geometry, Lift Generation & Pitching Moments

Syllabus Source: CAE Oxford (pp. 35-72) | Pass Target: >= 90% AviationExam

1. AIRFOIL ANATOMY & GEOMETRIC TERMINOLOGY

An airfoil is an aerodynamic cross-section engineered to generate usable lift while minimizing drag:

- **Chord Line:** A straight reference line connecting the leading edge and trailing edge centers of curvature. Chord length (c) is the reference datum for aerodynamic dimensions.
- **Mean Camber Line (MCL):** A line equidistant from the upper and lower airfoil surfaces. If the MCL coincides with the chord line, the airfoil is **SYMMETRICAL**. If MCL lies above chord, it is **POSITIVELY CAMBERED**.
- **Maximum Camber:** The maximum perpendicular distance between chord line and MCL, expressed as a percentage of chord (typically 2% to 4% at 30% to 40% chord). Camber determines zero-lift angle of attack.
- **Maximum Thickness:** The maximum distance between upper and lower surfaces (t/c ratio, e.g. 12% to 15% for transport aircraft, 9% to 11% for high-speed transonic airfoils).
- **Angle of Attack (Alpha):** The acute angle between the airfoil chord line and the **RELATIVE AIRFLOW** (freestream vector). Alpha is completely independent of the aircraft's pitch attitude relative to the horizon!

2. LIFT GENERATION: CIRCULATION, DOWNWASH & PRESSURE DISTRIBUTION

Lift is generated through two complementary aerodynamic mechanisms:

- **Pressure Differential (Bernoulli):** Air flowing over the curved upper surface accelerates, causing static pressure to drop below ambient (creating suction / low pressure). Air on the lower surface experiences slight compression. Upper surface suction contributes 70% to 80% of total aerodynamic lift!
- **Newton's Third Law (Downwash Deflection):** The airfoil deflects oncoming air downwards (downwash angle). The reaction to accelerating air downwards is an equal and opposite upward force: $LIFT = Downward\ Mass\ Flow \times Vertical\ Acceleration$.

3. CENTER OF PRESSURE (CP) VS AERODYNAMIC CENTER (AC)

Understanding the distinction between Center of Pressure and Aerodynamic Center is critical for aircraft stability:

Aerodynamic Datum	Physical Definition & Location	Behavior with Angle of Attack Changes
Center of Pressure (CP)	Point along chord where total resultant aerodynamic force acts.	MOVES with Angle of Attack! On a positively cambered airfoil: As AOA increases, CP MOVES FORWARD toward leading edge (up to stall). As AOA decreases, CP moves REARWARD. Inherently unstable!
Aerodynamic Center (AC)	Point along chord about which the pitching moment coefficient remains CONSTANT regardless of AOA.	DOES NOT MOVE with AOA in subsonic flight! Located at approximately the 25% CHORD POINT (Quarter-Chord) for all subsonic airfoils (moves to 50% chord in supersonic flight).

[!] EXAM TRAP (AVIATIONEXAM) - Pitching Moment About the Aerodynamic Center

- Symmetrical Airfoil: Pitching moment about the AC is ZERO at all angles of attack!
 - Positively Cambered Airfoil: Generates a NOSE-DOWN (negative) pitching moment about the AC that remains CONSTANT as AOA changes!
 - Center of Pressure Movement: On cambered wings, increasing AOA moves CP FORWARD; on symmetrical wings, CP remains virtually fixed at 25% chord.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): CP vs AC Identification**

Rule 1 (Aerodynamic Centre): If the question mentions 'constant pitching moment coefficient independent of angle of attack' or '25% chord subsonic', the answer is ALWAYS the Aerodynamic Centre (AC).
 Rule 2 (Centre of Pressure): If the question mentions 'point where total aerodynamic lift acts' or 'moves forward with increasing angle of attack on cambered airfoils', the answer is the Centre of Pressure (CP).
 Rule 3 (Symmetric Airfoil): On a purely symmetrical airfoil, CP and AC coincide at 25% chord and CP does not move within normal AoA.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: For a positively cambered subsonic airfoil, how does the Center of Pressure (CP) move when the Angle of Attack (AOA) increases?**

- [A]: CP moves rearward toward the trailing edge
- [B]: CP moves FORWARD toward the leading edge
- [C]: CP remains fixed at 50% chord
- [D]: CP moves downward

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. On positively cambered airfoils, increasing AOA intensifies leading-edge suction peak, drawing the resultant Center of Pressure forward toward the quarter-chord point.

Q2: What is the defining characteristic of an airfoil's Aerodynamic Center (AC) in subsonic flight?

- [A]: Lift is always zero at the AC
- [B]: The pitching moment about the AC remains CONSTANT as angle of attack changes, located at ~25% chord
- [C]: Total drag is zero at the AC
- [D]: It moves forward with speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Aerodynamic Center is the unique theoretical point where pitching moment coefficient is independent of AOA. In subsonic flow, it is fixed at approximately 25% chord.

Q3: What contributes the greatest percentage of total aerodynamic lift generated by an aircraft wing?

- [A]: High pressure pushing up against the lower surface (20-30%)
- [B]: Low static pressure (suction) acting on the UPPER surface (70-80%)
- [C]: Engine exhaust thrust

- [D]: Propeller wash

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Upper surface low pressure (suction) accounts for approximately 70% to 80% of total lift; lower surface positive pressure contributes only 20% to 30%.